Model-specific Local Feature Importance Methods for DNNs

Feature Importance

Lecture: Explainable Artificial Intelligence, ST 2026

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Outline

Main Directions

- Using *intermediate outputs*: Activation-map based
- Using *information flow*: Gradient- and back-propagation-based

Outline

Recap on DNNs

Using Activations

- Class Activation Mapping
- CAM Extensions

Using Gradients

- Vanilla Gradients
- Layerwise Relevance Propagation
- Smoothing or Integrating Gradients

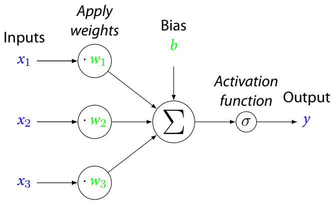
Outlook

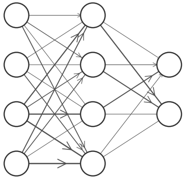
Summary

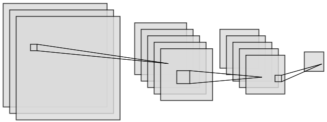
Recap on DNNs

A deep neural network (DNN) is concatenation of simple, differentiable functions (*neurons*). Main properties we will use here:

- organized in **layers** $\Rightarrow$ intermediate output vectors
- In many architectures, the intermediate output subdivides into subvectors (**pixels**) that are approximately aligned with the input,
  e.g., in convolutional deep neural networks (CNNs), patch-based vision transformers$\Rightarrow$ activation map pixels / patches
- neurons are differentiable $\Rightarrow$ **easy to differentiate** using chain rule.



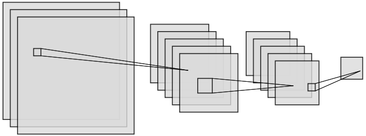




Outline

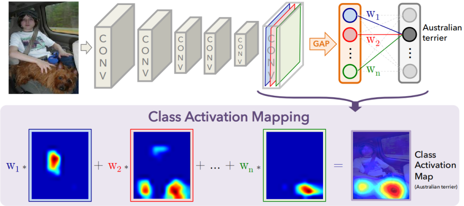
Class Activation Mapping

Idea: An input region can **only contribute** to the output, if the **pixel(s)** associated with it are **non-zero**.



1. record the *activation maps* of a layer
2. use their weighted sum as heatmap.

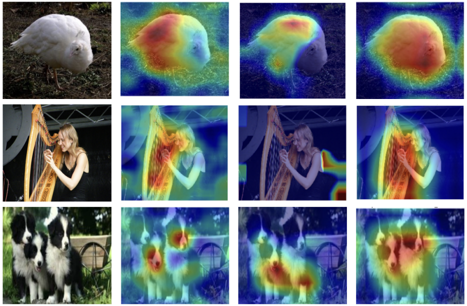
CAM [11]:
Use weights connecting activation maps to final output (after global average pooling operation)



CAM Extensions

Examples for different weighting schemes:

- Vanilla class activation mapping (CAM) [11]: connection **weights** to output
- Grad-CAM [10]: aggregated **gradient** of output wrt. activation map (the act. map's relevance)
- Grad-CAM++ [3]: Only **positive partial derivatives** as weights
- Eigen-CAM [9]: project activation map pixels to their **most principle component**
- SIDU [8]: **perturbation-based relevance** of activation maps & similarity of each activation map to the input



(a) Original Image (b) RISE (c) GRAD-CAM (d) SIDU

Pros and Cons

+ Very cheap & fast (single forward pass)
+ Causally motivated
– Fundamentally relies on spatial alignment between input and intermediate activation maps:
  - Flawed if the *receptive field* is too big
  - Very model specific!
– Highly depends on the weighting scheme
– Very coarse heatmaps

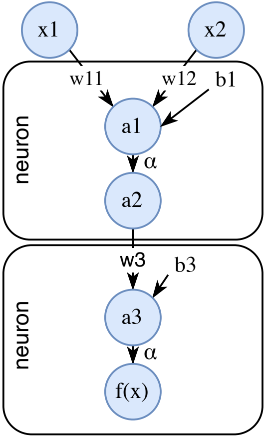
Outline

Recap: Chain Rule

Example (Chain Rule)

$$
\begin{aligned}
a_1 &= n_1(x) = w_{1,1}x_1 + w_{1,2}x_2 + b_1 \\
a_2 &= \alpha(a_1) \\
a_3 &= n_3(a_2) := w_2 a_2 \\
f(x;w) &= \alpha(a_3) = \alpha(n_3(\alpha(n_1(x_1,x_2))))
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial f}{\partial x_1} &= \frac{\partial \alpha}{\partial a_3}(a_3) \cdot \frac{\partial n_3}{\partial x_1}(a_2) \\
= \ldots &= \frac{\partial \alpha}{\partial a_3}(a_3) \cdot \frac{\partial n_3}{\partial a_2}(a_2) \cdot \frac{\partial n_2}{\partial a_1}(a_1) \cdot \frac{\partial n_1}{\partial x_1}(x)
\end{aligned}
$$

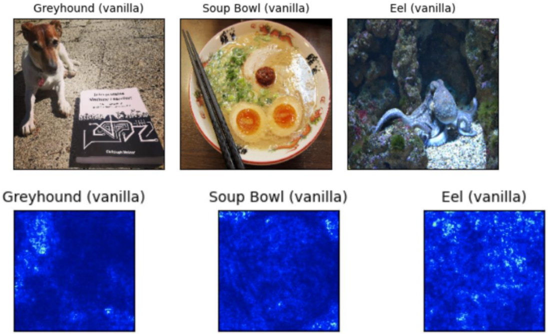


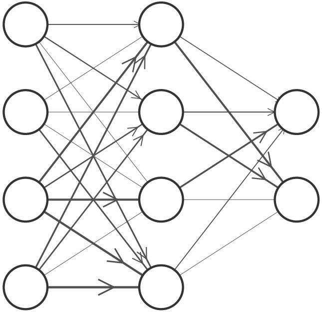
$\Rightarrow$ DNNs readily give rise to gradients w.r.t. weights and/or inputs!

Using Gradients

Vanilla *saliency* [1] (= gradient) calculation for model f on input x:

1. **Forward** pass through f:
   - Collect intermediate outputs *(a_i in example)*
   - Determine prediction $f(x)$, select to-be-explained output k.
2. **Backward** pass from k through f: Calculate $\frac{\partial f_k}{\partial x_i}(x)$ *using the a from the forward pass*.
3. **Visualize** positive, negative, or absolute values.





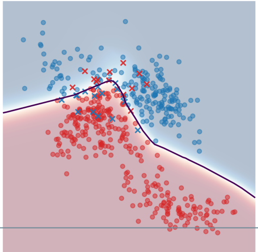
Limitations of Vanilla Saliency

- Vanishing gradients: What is a high saliency?
- Saturation: Once a class probability is high, increasing the input will not increase the class probability too much (global importance becomes invisible).

Example (Saturation)

Increasing the income feature for a person classified as rich does not increase the (already high) confidence that they are rich.

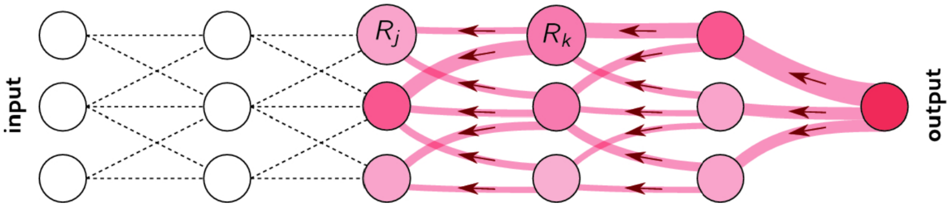
- Instability: the decision boundary of classical DNNs is not smooth and has locally fluctuating gradients.



Layerwise Relevance Propagation

Idea: Backpropagate relevance from output to input (like the gradient); avoid vanishing gradients by demanding *completeness* of neuron relevances R_i:

$$\forall \text{layer } l, l': \sum_i R_{l,i} = \sum_i R_{l',i} \quad \text{(total relevance stays constant over layers)}$$



The curse of choice

Example propagation rules fulfilling this:

- **LRP-0**: $R_k^{(l)} = \sum_i \frac{\overbrace{a_k w_{k,i}}^{\text{contribution of } k \text{ to relevance of } i}}{\sum_j a_j w_{k,j}} R_i^{(l+1)}$
- **LRP-ϵ**: $R_k^{(l)} = \sum_i \frac{a_k w_{k,i}}{\epsilon + \sum_j a_j w_{k,j}} R_i^{(l+1)}$ (more stable)
- **LRP-γ**: $R_k^{(l)} = \sum_i \frac{a_k (w_{k,i} + \gamma w_{k,i}^+)}{\sum_j a_j (w_{k,j} + \gamma w_{k,j}^+)} R_i^{(l+1)}$ (prefers positive)

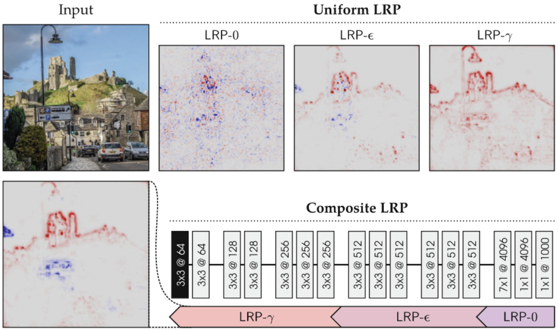
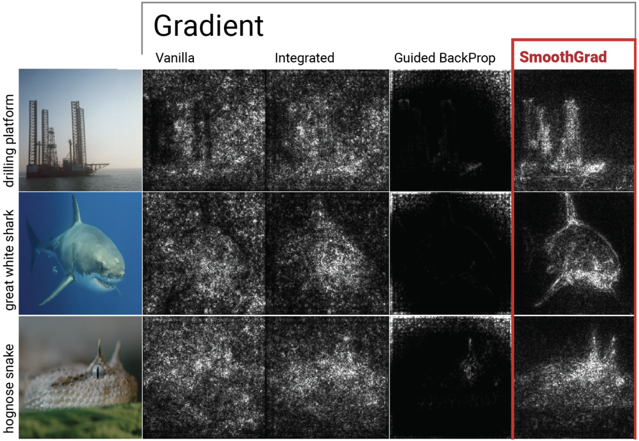


Fig. 10.4. Input image and pixel-wise explanations of the output neuron 'castle' obtained with various LRP procedures. Parameters are $\epsilon = 0.25$ std and $\gamma = 0.25$.

Tackling Instability Directly: SmoothGrad

Idea: Generate multiple version $x' = x + \epsilon$ of x by adding noise ϵ to it, then **average pixel attributions.**



But this is expensive!

Tackling Saturation: Integrated Gradients from a Baseline

Idea: Instead of sampling locally around a sample like SmoothGrad, average gradient evolvement between a **baseline** x_0 and x:

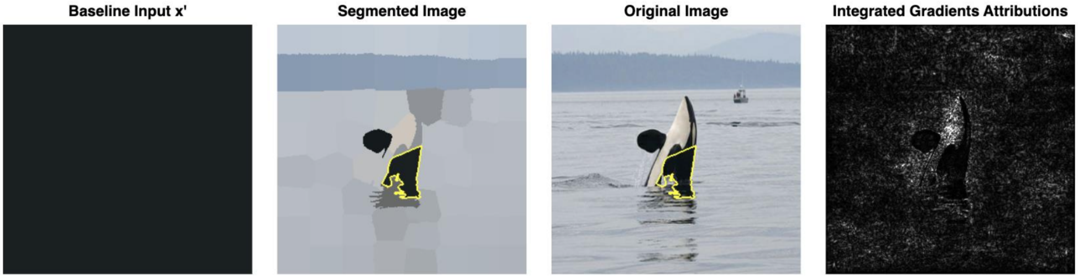
$$\mathrm{IG}_i(f, x, x_0) = \overbrace{(x - x_0)}^{\text{Diff. from baseline}} \cdot \underbrace{\int_{\alpha=0}^{1}}_{\text{From baseline to input ...}} \overbrace{\frac{\partial f(x_{0,i} + \alpha(x_i - x_{0,i}))}{\partial x_i} d\alpha}^{\text{...accumulate local gradients}}$$

$$\approx (x - x_0) \cdot \sum_{\alpha \in [0, s, 2s, \ldots, 1]} \frac{\partial f(x_{0,i} + \alpha(x_i - x_{0,i}))}{\partial x_i} \quad \text{for a step-width } s$$

satisfies *completeness* (= no vanishing gradients): $\sum_i \mathrm{IG}_i(f, x, x_0) \overset{\overset{\text{quantifies approximation error}}{\downarrow}}{=} f(x) - f(x_0)$

Baseline Choice for Integrated Gradients

the choice of baseline must be carefully made and is highly task dependent



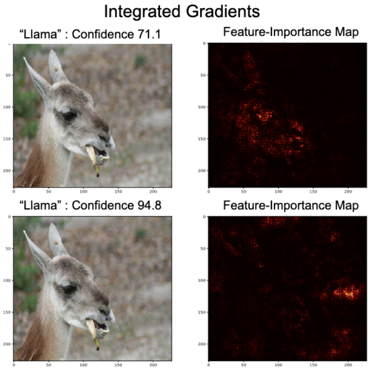
A dark baseline yields small integrated gradients of dark pixels: The black parts of the orca are not highlighted in the explanation.

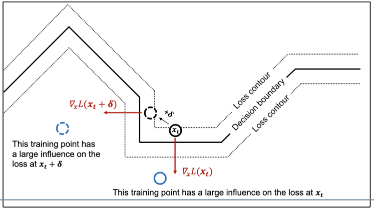
(Averaging across multible baselines gets very costly.)

Outline

Outlook

- ▶ Integrated Hessians [6]: White-box calculation of feature interactions
- ▶ Combining heatmaps: Just multiply!
- \+ Heatmaps can be used for training regularization (*Jacobian regularization* [2])
- – Explanations can be maliciously attacked [5]: "we demonstrate how to generate adversarial perturbations that produce perceptively indistinguishable inputs that are assigned the *same* predicted label, yet have very *different* interpretations."





Outline

Summary

- Insights used from DNNs for white-box feature importance explanations:
  - intermediate outputs (activation maps)
  - information flow (gradient)
- Principled techniques: layerwise relevance propagation (LRP) and Integrated Gradients

\+ White-box methods can generally be cheaper

– Gradient-based methods may suffer from saturation and instability; sampling to avoid this is costly

Further Reading

- Christoph Molnar, *Interpretable Machine Learning* [4], Chap. 28
- Montavon et al., "Layer-Wise Relevance Propagation" [7] for LRP
- The captum framework implements many of the discussed feature importance methods. Check out their documentation on vanilla saliency, Integrated Gradients, (guided) Grad-CAM, and LRP.

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[1] David Baehrens et al. "How to Explain Individual Classification Decisions". In: *Journal of Machine Learning Research* 11 (Aug. 2010), pp. 1803–1831. ISSN: 1532-4435.

[2] Alvin Chan et al. "Jacobian Adversarially Regularized Networks for Robustness". In: *International Conference on Learning Representations*. Sept. 2019. (Visited on 07/05/2023).

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[4] Christoph Molnar. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. 3rd ed. 2025.

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[10] Ramprasaath R. Selvaraju et al. "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization". In: *Proc. 2017 IEEE Int. Conf. Computer Vision*. Venice: IEEE, Oct. 2017, pp. 618–626. ISBN: 978-1-5386-1032-9. DOI: 10.1109/ICCV.2017.74. (Visited on 11/14/2020).

References II

[11] Bolei Zhou et al. "Learning Deep Features for Discriminative Localization". In: *Proc. 2016 IEEE Conf. Comput. Vision and Pattern Recognition*. Las Vegas, NV, USA: IEEE Computer Society, 2016, pp. 2921–2929. DOI: 10.1109/CVPR.2016.319.